Materials Science 1 – Recovery, recrystallisation and grain growth

Learning objectives and Learning outcomes

- Understanding how to reverse the effects of cold working;

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- Introduce the process of annealing materials for improving properties;
- Interpret data from annealing processes;
- Estimate the size of grains after recovery.

Effects of cold working

By plastically deforming polycrystalline metals at temperatures much below their melting temperature produces the following microstructural changes:

(1) a change in grain shape;
(2) strain hardening, 应变硬化
(3) an increase in dislocation density.

Some fraction of the energy spent to deform the metal is stored in the metal as strain energy, associated with tensile, compressive, and shear zones around newly created dislocations.

Effect of Heat Treatment After Cold Working

Some of the microstructural changes and strain caused by cold working can be reverted by appropriate heat treatments, often termed "**annealing treatments**".

Annealing

A heat treatment in which a material is exposed to an elevated temperature for an extended time period and then slowly cooled. Any annealing process consists of three stages:

(1) heating to the desired temperature,
(2) holding or "soaking" or "dwelling" at that temperature, and
(3) cooling, usually to room temperature.

Annealing temperature and time are the important parameters in these procedures.

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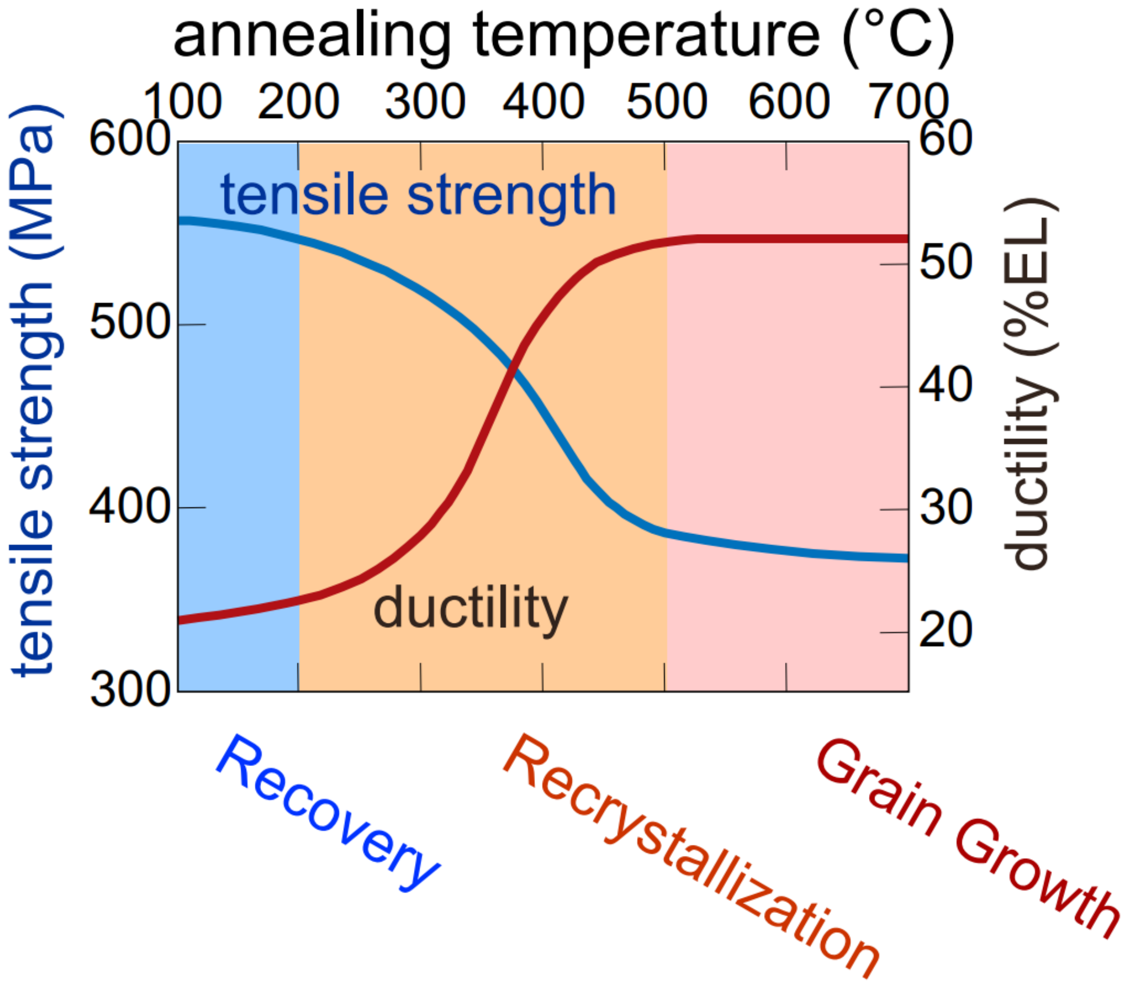
Annealing is carried out to:
(1) relieve stresses;
(2) increase softness, ductility, and toughness;
(3) produce a specific microstructure.

Effect of Heat Treatment After Cold Working

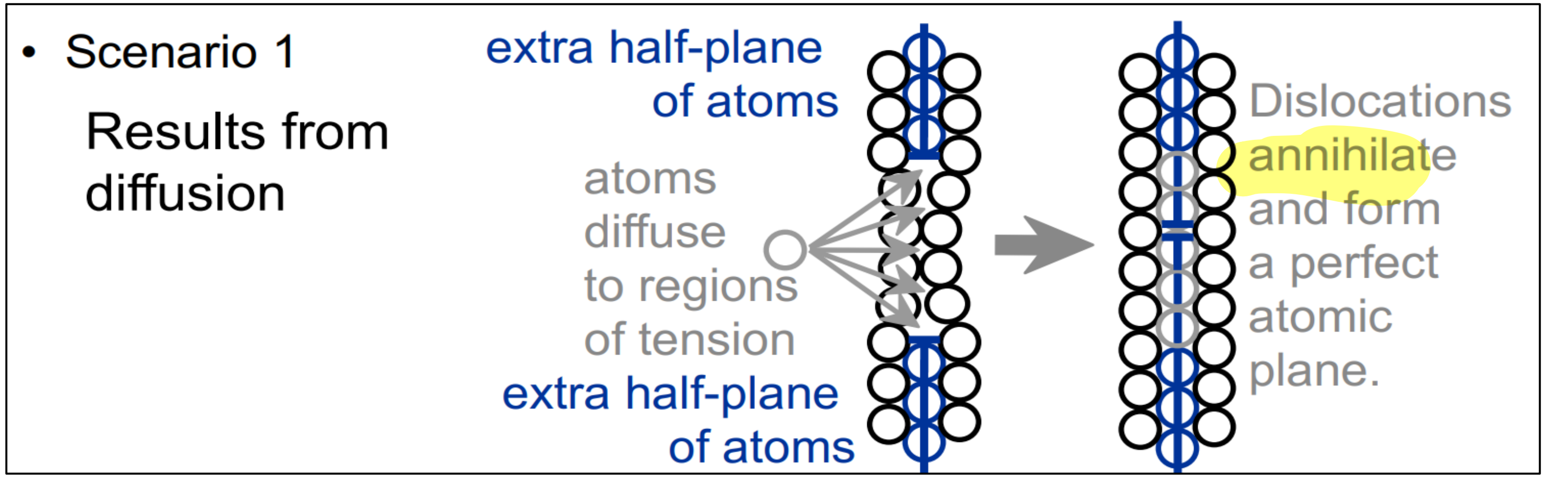
Annealing treatments after cold working usually eliminate the effects of the cold working. The process occurs in three main steps: **1. Recovery; 2. Recrystallization; 3. Grain growth.**

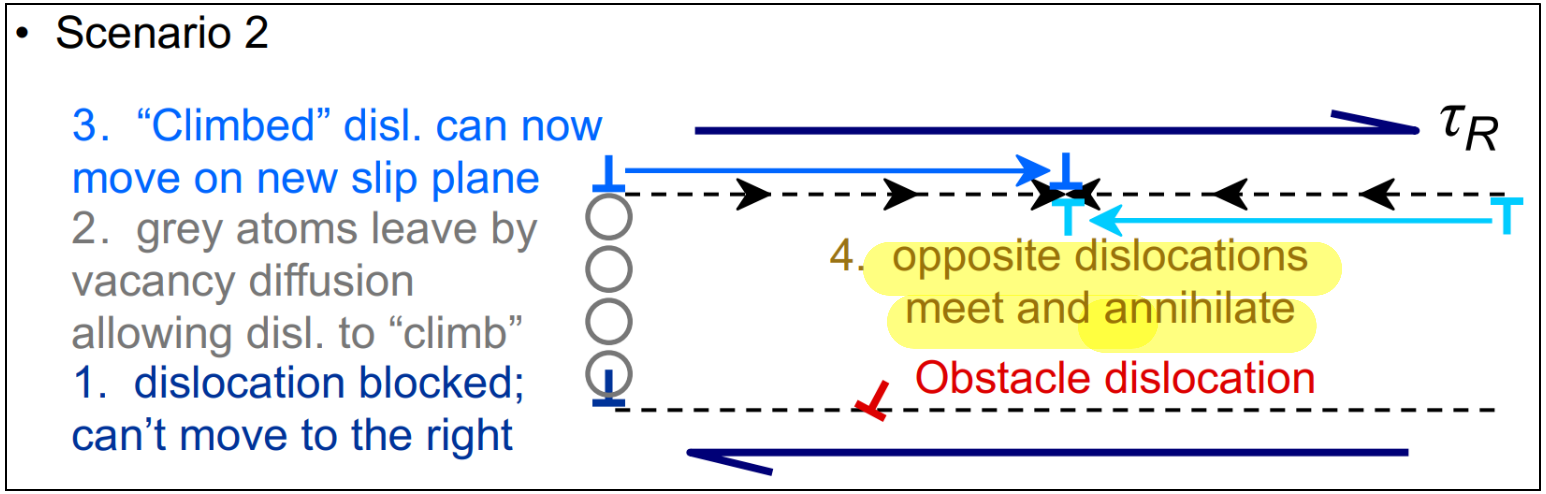


Annealing treatments usually results in a reduction of tensile strength and an increase in ductility.

Recovery

During recovery, some of the stored internal strain energy is relieved through the reduction of dislocation density. This may occur by two main processes: 1. Diffusion; 2. Dislocation climbing and annihilation.



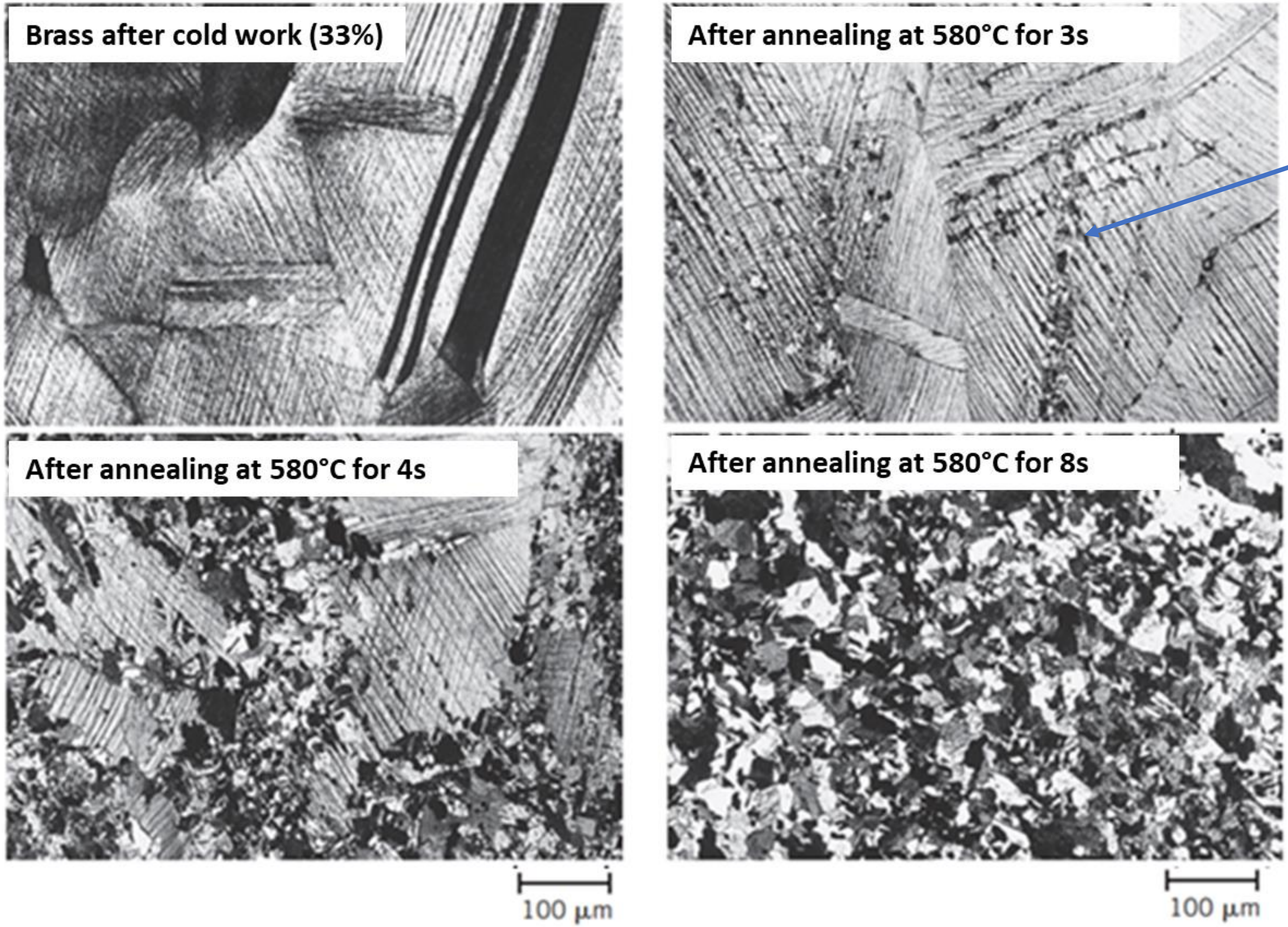


Recrystallization

- Formation of a new set of strain-free and equiaxed grains with low dislocation densities. This occurs by nucleation-growth process that involves short-range diffusion.
- Also, during recrystallization, the mechanical properties are restored to their original values, the metal becomes softer and weaker, but more ductile.
- The extent of recrystallization depends on time and temperature.
- The recrystallization temperature is typically between **one-third and one-half of the absolute melting temperature** of a metal or alloy and depends on several factors, including the amount of prior cold work and the purity of the alloy.
- Increasing the percentage of cold work lowers the recrystallization temperature and enhances the rate of recrystallization.
- Below a critical degree of cold work (2-20%), recrystallization does not occur this is between.
- Recrystallization proceeds more rapidly in pure metals than in alloys (impurities stops grain growth).

Recrystallization

Example of recrystallization on cold worked brass



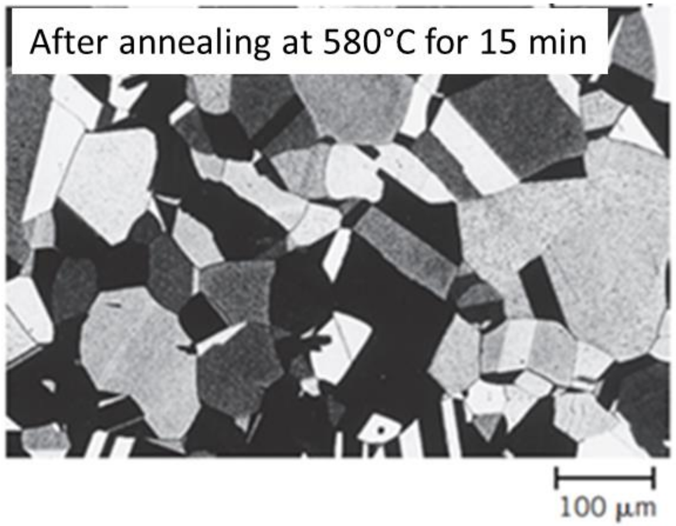
Formation of new nuclei

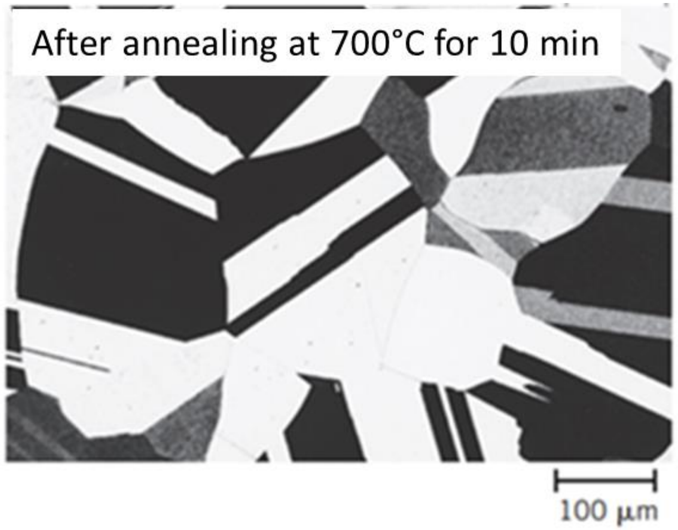
Amount of new nuclei increases with time.

Grain growth

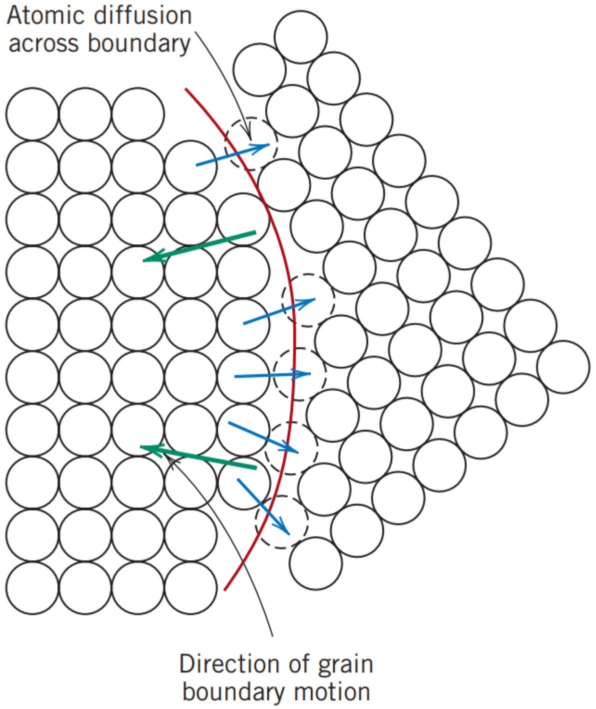
At longer times, average grain size increases:

- Small grains shrink and ultimately disappear;
- Large grains continue to grow.





Grain growth occurs by the migration of grain boundaries due to short-range diffusion of atoms from one side of the boundary to the other. The direction of grain boundary motion and atomic diffusion are opposite to each other.



Grain growth

Grain growth is a thermally activated phenomenon. It is based on particles diffusion. The grain-growth mechanisms can be elucidated by estimating the activation energy for grain growth.

$$\boxed{\frac{dD}{dt} = \frac{G}{D}}$$

where

$$\boxed{G = G_0 \exp\left(-\frac{E_g}{RT}\right)}$$

G_0: pre-exponential term;
E_g: activation energy;
R: gas constant;
T: absolute temperature.

Integrating:

$$\boxed{D^2 - D_0^2 = \left[2G_0 \exp\left(-\frac{E_g}{RT}\right)\right] t}$$

D: grain size at time t;
D_0: grain size at time t=0;
t: time.

This equation is normally valid for high-purity materials and in absence of inhibiting effects to grain growth. When inhibiting effects are significant, grain growth can be usefully described by the following empirical expression:

$$\boxed{D^n - D_0^n = \left[2\bar{G}_0 \exp\left(-\frac{E_g}{RT}\right)\right] t}$$

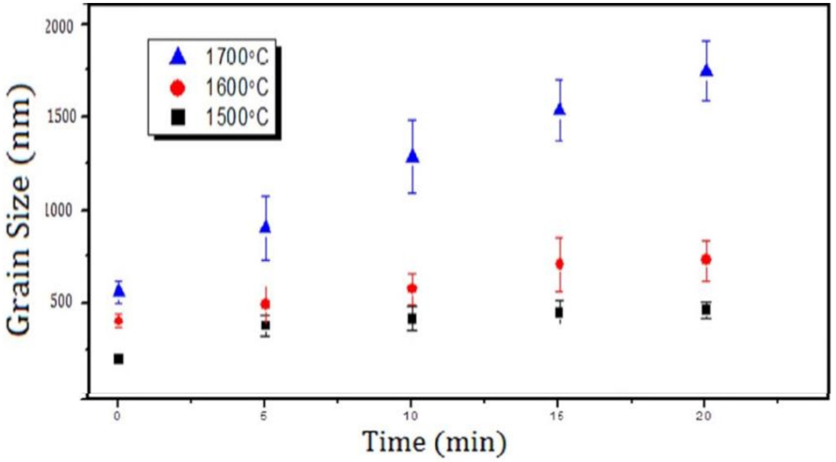
n: grain growth exponent (usually ≥2). It depends on the drag mechanisms, related to inclusions, impurities or grain growth inhibitors.

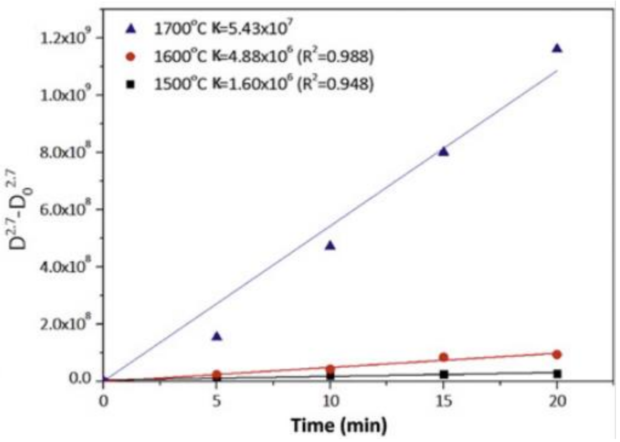
Grain growth

The previous can be written as:

$$\boxed{D^n - D_0^n = Kt}$$

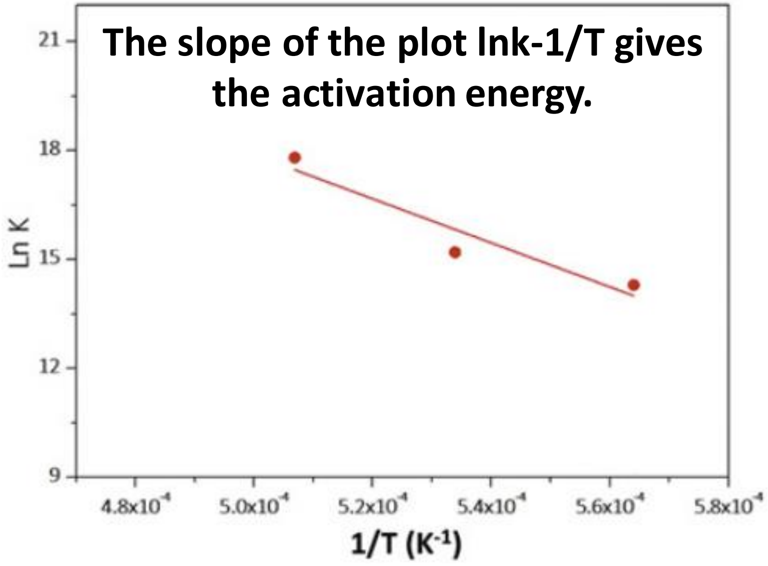
The grain growth exponent and the term related to the grain growth rate K, can be obtained from isothermal grain growth curves at different temperatures.





Once K is obtained, the activation energy of grain growth can be obtained by using the relationship:

$$\boxed{K = B\,exp\left(-\frac{E_g}{RT}\right)}$$



Grain growth

Example 1: Grain growth.

The average grain diameter for a brass material was measured as a function of time at 650°C, which is shown in the following table at two different times.

Time (min)	Grain diameter (mm)
40	5.6×10^{-2}
100	8.0×10^{-2}

Assuming normal grain growth (n=2):

(a) What was the original grain diameter?

(b) What grain diameter would you predict after 200 min at 650°C?

Grain growth

Example 1 solution

a) Original grain diameter $d^2 - d_0^2 = Kt$

$$(5.6 \times 10^{-2}\ \text{mm})^2 - d_0^2 = (40\ \text{min})K$$

$$(8.0 \times 10^{-2}\ \text{mm})^2 - d_0^2 = (100\ \text{min})K$$

Solution of these expressions yields a value for $d0$ = 0.031 mm

and a value for K of $K = 5.44 \times 10^{-5}\ \text{mm}^2/\text{min}$

Grain growth

Example 1 solution

(b) At 200 min, the diameter d is calculated by re-arranging the grain diameter equation and incorporating values of d_0 and K as follows:

$$d = \sqrt{d_0^2 + Kt}$$

$$= \sqrt{(0.031\,\text{mm})^2 + \left(5.44 \times 10^{-5}\,\text{mm}^2/\text{min}\right)(200\,\text{min})}$$

$$= 0.109\ \text{mm}$$

Summary

Plastic deformations induce:

- changes in grain shape;
- strain hardening;
- Increase in dislocation density.

These effects can be reverted back to the precold- worked states by appropriate heat treatment

Restoration results from two different processes that occur at elevated temperatures:

- recovery;
- recrystallization.

Longer annealing treatments after recrystallization will induce grain growth, which is a thermally activated phenomenon based on diffusion of atoms.